

CD-A OGS Inversion Blocker Discussion Packet

Prepared as a self-contained discussion and request packet

Generated 2026-06-01

Purpose

This packet puts the remaining external questions, provider file requests, and modelling-team decisions in one place for discussion. It is intended to be sent as a standalone document: all questions, required evidence, context, and response instructions needed by the recipients are included here.

Important email-safety note: this document does not send email. The mailbox state checked on 2026-06-01T03:55:46+02:00 recorded six prepared request emails, all still unsent. The check found zero sent copies of the generated request subjects, zero provider-reply hits, and zero recent non-draft CD-A/HERMES/TeamBeam hits in the searched scope.

How recipients should respond: reply by email with the request id(s) in the subject, body, or filename. Attach machine-readable source tables, scripts, calibration notes, geometry files, or provenance notes directly to the reply where possible. For every response, include source contact, response date, units, coordinate/reference frame, processing provenance, quality/status flags, and uncertainty/covariance assumptions where applicable. No access to local folders or previous file paths is required.

Current Status Snapshot

Item	Current state
Overall objective readiness	not_complete . The non-complete requirements are the permeability field/final combined inversion and remaining open questions.
Final promotion decision	do_not_promote_current_field . The current field is an active-objective incumbent only.
Open blockers	8 total: 7 measurement/provider gates and 1 model-provenance confirmation.
Request email state	6/6 expected request emails prepared, 6 still unsent, no sent-subject hits, no provider-reply hits.
Recovered material state	Email, file-transfer, and download recovery scans found zero additional gate-closing rows.
Report and catalogue state	Active report comments resolved; citation locators pass; measurement traceability passes; frozen-model inclusion audit passes.

High-Level Discussion Decisions

1. [] Decide whether to send the six request emails as written, edit them first, or route them differently.
2. [] Decide whether to wait for provider responses or explicitly mark some gated streams as diagnostic-only/excluded for the final objective.
3. [] Decide the final NMR residual semantics.
4. [] Decide the direct-permeability likelihood/support/outlier policy.
5. [] After responses or exclusions are recorded, rerun the stream gates, conditional scenarios, field-selection audit, final-promotion checklist, final objective decision register, and objective-readiness audit.

1 Open Blocker Register

Request id	Priority / stream	Email route	Missing evidence	What closes the gate
<code>ert_transform_support</code>	high / ERT resistivity	Request email 1; To Gesa; Cc Markus Furche	ERT-to-OGS transform and exact near-niche support mask are assumed, not confirmed.	Coordinate-frame note or transform script, coordinate reference, accepted tunnel/niche support polygon or mask, and decision on 35 cm rock-depth handling.
<code>ert_uncertainty</code>	high / ERT resistivity	Request email 1	No defensible ERT residual sigma/correlation model; dense VTK pixels cannot be treated as independent rows.	Per-cell, regional, or time-level uncertainty/covariance export, or a simplified weighting rule with log base, filtering, spatial correlation, and effective degrees of freedom.
<code>hm_numeric_exports</code>	high / other HM monitoring	Request email 2; To Gesa	Available collected files contain layout/qualitative evidence, but no hard-residual-ready Geoscope time series, laser statistical product, or full levelling table.	Machine-readable Geoscope mini-piezometer, extensometer, crackmeter, laser-scan interpretation, and levelling tables with time/epoch and support.
<code>hm_uncertainty</code>	high / other HM monitoring	Request email 2	Metadata for hard HM residual weights are absent.	Unit/reference convention, uncertainty model, covariance/independence assumptions, quality flags, failure periods, laser registration uncertainty/masks, and levelling covariance frame.
<code>rh_active_curve_provenance</code>	high / RH/suction	Request email 3; To Gesa	Active boundary-curve generation, sensor screening, time axis, constants, and extension policy are not confirmed.	Source table/script, sensor selection note, Kelvin conversion constants, model-time origin/timezone, smoothing/manual-edit policy, sign convention, and open/closed mapping.
<code>taupe_unit_calibration</code>	high / Taupe/TDR	Request email 4; To Gesa	Taupe values are not confirmed as volumetric water-content percent, permittivity, or another ARDP-derived proxy.	Workbook unit definition, sensor/band calibration equations, uncertainty by sensor/band, baseline/reference date, and dielectric/water-content conversion notes.

Request id	Priority / stream	Email route	Missing evidence	What closes the gate
perm_endpoint_geometry	medium / permeability pulse tests	Request email 5; To Gesa	Historical permeability endpoints are needed only if older rows should enter the fit.	Endpoint coordinates, digitized traces, or interval geometry tables for BCD-A24, BCD-A25, BCD-A26, BCD-A27, and BFM-D19 intervals.
cte_value_confirmation	high / model provenance	Request email 6; To Gesa; Cc Tuanny	The XML binds CTE to solid thermal expansivity, but the value/comment look like a copied heat-capacity entry.	Provider confirmation that the value is intended, inactive/irrelevant, a typo not to be changed in the frozen model, or another documented provenance case.

2 Provider Request Email Pack

Request Email 1: ERT transform, support, and uncertainty

Subject	CD-A ERT transform, support, and uncertainty needed for OGS comparison
To / Cc	Gesa.Ziefle@bgr.de / Markus.Furche@bgr.de
Contact route	Coordinator with named provider Cc. Markus Furche is found in ERT/meeting context, but no direct Gmail sender messages from Markus were found in the scan.

Questions to ask / files requested

- Confirm the ERT coordinate origin, axes, units, and the currently assumed transform `model_x=raw_x`, `model_y=raw_y+500`.
- Provide or confirm the accepted near-niche support mask/polygon used for comparing ERT to OGS.
- State how 35 cm rock-depth handling and radial support variants should be treated.
- Provide the ERT uncertainty model: sigma units/log base, time grouping, spatial correlation, filtering rules, unstable-cell handling, and whether cells may be aggregated before comparison.

Acceptance evidence

- ERT spatial projection/support sensitivity can be regenerated without a provisional-transform caveat.
- ERT can be weighted without treating dense VTK cells as independent residual rows.

Requested response package: reply by email with request id `ert_transform_support` and/or `ert_uncertainty`. Attach source tables, scripts/notebooks, polygons/masks, coordinate-frame notes, and uncertainty/covariance descriptions directly to the reply.

Request Email 2: Geoscope, laser-scan, levelling, and other HM exports

Subject	CD-A Geoscope, laser-scan, and levelling numeric exports needed for model validation
To / Cc	Gesa.Ziefle@bgr.de / none
Contact route	Coordinator only. Ask Gesa to forward to the relevant BGR Geoscope, laser-scan, and levelling data owners if needed.

Questions to ask / files requested

- Machine-readable Geoscope mini-piezometer time series.
- Extensometer and crackmeter time series if available.
- Laser-scan statistical interpretation products, not only surface geometry.

- Full precision-levelling tables, including survey epochs and reference convention.
- For every numeric export: instrument id, timestamp/epoch, value, unit, coordinate/support id, reference/zero convention, processing provenance, quality/status flag.
- For every candidate residual: uncertainty/covariance or an accepted simplified weighting rule, including failed-sensor and maintenance intervals.

Acceptance evidence

- At least one other-HM request class becomes hard-residual-ready with time/epoch and model-facing support.
- Each residual has enough metadata to state what OGS quantity/support it measures and what residual weight it receives.

Requested response package: reply by email with request id `hm_numeric_exports` and/or `hm_uncertainty`. Attach machine-readable tables and metadata directly to the reply, preferably one file per instrument class plus a short provenance/readme note.

Request Email 3: RH/suction boundary-curve provenance

Subject	CD-A RH/suction boundary-curve provenance needed for OGS forcing
To / Cc	Gesa.Ziefle@bgr.de / none
Contact route	Coordinator only; no separate RH/OGS pressure-curve processing owner was identified in the scan.

Questions to ask / files requested

- Active open-niche pressure-curve source table and any script/notebook used to generate the XML curve.
- Sensor selection/screening note, including failed or excluded sensors.
- Model-time zero, timezone, RH percent/fraction convention, pressure unit/sign convention, and temperature/density constants.
- Kelvin conversion constants and any smoothing/manual-edit or extrapolation policy.
- Open/closed curve mapping and post-active-curve extension policy.

Acceptance evidence

- The active XML pressure curve can be regenerated or explained well enough that replacement curves/retention targets are no longer based on unknown provenance.

Requested response package: reply by email with request id `rh_active_curve_provenance`. Attach the source table, generation script/notebook, screening note, and any pressure/RH conversion assumptions directly to the reply.

Request Email 4: Taupe/TDR units and calibration

Subject	CD-A Taupe/TDR workbook units and calibration needed before objective activation
To / Cc	Gesa.Ziefle@bgr.de / none
Contact route	Coordinator only. Ask Gesa to forward to the Taupe/TDR provider or confirm the source directly.

Questions to ask / files requested

- Define the units and meaning of `Taupe_WC.xlsx` values by sheet/column.
- Confirm whether values are volumetric water-content percent, permittivity, or another ARDP-derived proxy.
- Provide sensor/band calibration equations, constants, baseline/reference date, and uncertainty model.
- State whether matrix/porosity corrections have already been applied.

Acceptance evidence

- The Taupe semantics audit can decide whether absolute Taupe values are water content, permittivity/proxy, or trend-only evidence.

Requested response package: reply by email with request id `taupe_unit_calibration`. Attach the work-book unit definition, calibration equations/constants, baseline/reference information, uncertainty model, and any matrix/porosity correction notes directly to the reply.

Request Email 5: Historical permeability endpoint geometry

Subject	CD-A historical permeability endpoint geometry needed for inactive intervals
To / Cc	Gesa.Ziefle@bgr.de / none
Contact route	Coordinator only. Ask Gesa to forward to the BGR permeability data owner if interval endpoints come from another source.

Questions to ask / files requested

- Endpoint coordinates, labelled digitized traces, or interval geometry tables for historical BCD-A24, BCD-A25, BCD-A26, BCD-A27, and BFM-D19 permeability intervals.
- For every interval: borehole id, open/closed assignment, start/end coordinates or depths, orientation convention, interval length/support, source date/table, permeability value, uncertainty/evaluation note.
- If endpoint geometry will not be used, confirm whether these older rows should remain inactive/excluded from the final fit.

Acceptance evidence

- Inactive historical rows can be projected to OGS cells with a trace/support definition rather than excluded for missing geometry.

Requested response package: reply by email with request id `perm_endpoint_geometry`. Attach endpoint coordinate tables, digitized traces, interval-geometry files, or a written confirmation that the older rows should remain inactive/excluded from the final fit.

Request Email 6: CTE value/provenance confirmation

Subject	CD-A OGS CTE value/provenance confirmation needed before thermal-parameter interpretation
To / Cc	Gesa.Ziefle@bgr.de / Tuanny.Cajuhi@bgr.de
Contact route	Coordinator with model-setup Cc.

Questions to ask / files requested

- Confirm whether the XML CTE value bound to solid thermal expansivity is intended.
- If not intended, state whether it is inactive/irrelevant for the current run, a known typo not to be changed in the frozen model, or copied from another parameter such as heat capacity.
- Provide the provenance/source document or model-setup note for the value.

Acceptance evidence

- The report open-comment audit can move the CTE row from confirmation-required to confirmed inactive, confirmed typo, or confirmed value with the provider response cited.

Requested response package: reply by email with request id `cte_value_confirmation`. Include the intended interpretation of the current XML value and attach or cite the source/provenance note for that value.

3 Internal Modelling Decisions

NMR final residual semantics

Decision required before final promotion. The current report keeps `retain_raw_absolute_theta_current_report_default_with_caveats`. The recommended provisional candidate is `within_label_trend_anomaly`, but the acceptance template records 0/1 approvals.

Option	Current status	Discussion consequence
raw_absolute_current	active but conditional	Keeps the current active objective, but compares total NMR-visible water content to mobile OGS water content without a bound/interlayer-water correction.
global_uniform_offset	stress test only	Reject unless collaborators provide one physically justified global offset.
absolute_with_label_bias	conditional	Preserves absolute information but needs approved bias/offset priors and wording that biases are not transport water.
within_label_trend_anomaly	recommended after acceptance	Safest provisional NMR residual because constant bound/interlayer/campaign offsets cancel; absolute water content is not fitted.

Required signoff fields: approval status, primary policy selection, approver name, approval date, approval reference, and accepted residual definition.

Direct-permeability likelihood/support policy

Decision required before reopening same-support OGS spending or final field promotion. The current-report policy is `keep_rowwise_gaussian_default`. The same-support active-objective batch gate is closed because the current field is already at the support lower bound for the current support map.

Option	Diagnostic value	Discussion consequence
keep_rowwise_gaussian_default	269.818057	Reproducible current default, but top row losses and support conflicts must be explicitly accepted.
use_robust_row_likelihood	Student-t diagnostic 121.169	Needs a written kernel, scale, outlier treatment, and comparison rule against historical row-Gaussian objectives.
aggregate_by_model_support_cell	weighted-mean diagnostic 1.77×10^{-28}	Needs grouping key, statistic, group weights, and reporting of within-support observation ranges.
gate_configured_scalar_outliers	inside-only 245.675	Needs row-specific outlier disposition and explicit bounds/tensor-shape decision.
new_parameterization_before_more_ogs	gate option	Requires a named field family and materiality threshold before spending more OGS runs.

Support-conflict evidence to acknowledge: 30 active support cells, 24 repeated support cells, 16 cells spanning at least two log10 units, with the top conflict at cell 4648, BCD-A32, 0.85-0.87 m, observed range 6.94885 log10.

Final objective scenario selection

The current field wins only the current raw-NMR scenario and ties the direct-only scenario. Every currently scored scenario that includes promoted NMR, ERT, or Taupe selects a different field.

Scenario	Winner	Current rank	What is still required
F01_current_raw_nmr_exclude_gated_streams	Current field	1	Explicitly exclude or keep diagnostic-only all gated streams.
F02_promoted_nmr_trend_only	broad_continuous_001_003_length_0p021m	14	Approve NMR trend/anomaly policy.
F03_raw_nmr_plus_ert	local_basis_sampler_003_basis_029_det_1_0p0100_s_1p000	3	Close ERT transform/support and uncertainty gates.
F04_raw_nmr_plus_taupe	local_bracketed_003_length_0p031m	21	Close Taupe/TDR unit/calibration gate.
F05_raw_nmr_plus_ert_taupe	production_sampler_round3_002_length_0p028m_shift_1p050	19	Close ERT and Taupe/TDR gates.
F06_promoted_nmr_plus_ert_taupe	production_sampler_round3_002_length_0p028m_shift_1p050	18	Approve NMR policy and close ERT/Taupe gates.

Scenario	Winner	Current rank	What is still required
F07_rank_consensus_diagnostic_selection	local_basis_sampler_003_basis_029_det_1_0p0100_s_1p000	2	Record explicit waiver that diagnostic ranks, not likelihood, select the field.
F08_exclude_nmr_direct_permeability_only	Current field tie	1	Explicitly exclude NMR and decide direct-only tie-break.
F09_wait_for_rh_other_hm_endpoint_expansion	not scored	n/a	Wait for RH, other-HM, or endpoint evidence, then rebuild scenarios.

4 After Replies Arrive: Internal Follow-Up

1. Record which request ids were sent, which people received them, and which answers/files came back.
2. For every received answer, record source contact, response date, file names, units, coordinate/reference frame, processing provenance, quality/status flags, uncertainty/covariance assumptions, and which acceptance evidence it satisfies.
3. Save received files into the project archive under the matching measurement stream, preserving original file-names where possible and adding a short readme/provenance note.
4. Rerun the stream gates, conditional scenario matrix, field-selection audit, final-promotion checklist, final objective decision register, and objective-readiness audit in the working repository.
5. If new evidence changes the modelling interpretation, update the report text, citations, measurement traceability notes, and frozen-model inclusion audit before making any final field-promotion claim.

5 Packet Provenance

This document is a self-contained summary of the current CD-A OGS inversion blockers, provider questions, and modelling-team decisions. It intentionally does not require recipients to open local folders, internal tracker files, or previous email drafts. The request ids in this packet are the stable labels to use when discussing or replying to the open issues.